QUANTUM COMPUTING 2026

The Era of Quantum-Centric Supercomputing

Strategic Report 2026

EXECUTIVE SUMMARY

QCSC combines classical HPC, quantum processors, and AI.

Market projected to $11B by 2030.

1. ARCHITECTURAL SHIFT

QCSC relies on tightly integrated stacks with co-located GPUs and QPUs.

- QCSC: IBM reference architecture
- NVQLink: High-speed links for error correction
- Co-locality: Scale-up systems integration

2. BREAKTHROUGH ALGORITHMS (SQD)

Tryptophan Cage: 12,635 atoms simulated using QCSC

Infrastructure: 152,064 classical nodes + QPU

Metric	Value
Infrastructure	152K nodes + QPU
Speedup	100x base
Accuracy	AMole calc

3. AI-POWERED ERROR CORRECTION

NVIDIA Ising Decoder: 347.7x error reduction vs Chromobius

7.3x faster, 0.3% physical error rate

4. MARKET INSIGHTS

Value	Metric
$11B	Market by 2030
$125M	ORNL Investment
152K	HPC nodes
Qiskit	Software Standard

RECOMMENDATIONS

- Adopt HPC mindset with co-located resources
- Invest in NVIDIA H100/B200 infrastructure
- Focus on chemistry/materials science applications

VISUAL CAPABILITIES:

- IBM Reference Architecture Diagram
- Fusion Energy Applications